SEA4001W: Exercise 13 - The discretized Ekman equations

Tristan Mitchell: MTCTRI001

AI Declaration

I asked Claude.ai to help with the math on this exercise, but after understanding it, I rewrote it in a way that made sense to me.

P1

Exercise 13

eq 4.2

$$\frac{\partial u_E}{\partial t} - f v_E = A_v \frac{\partial^2 u_E}{\partial z^2}$$

$$\frac{\partial v_E}{\partial t} + f u_E = A_v \frac{\partial^2 v_E}{\partial z^2}$$

$$\left.\begin{aligned} z_K &= z_0 - K\Delta z \\ t_i &= t_0 + n\Delta t \\ z_0 &= 0 \end{aligned}\right\} \text{given on slide 146}$$

$$\frac{\partial u_E}{\partial t} \approx \frac{u_K^{n+1} - u_K^n}{\Delta t} \qquad \frac{\partial v_E}{\partial t} \approx \frac{v_K^{n+1} - v_K^n}{\Delta t}$$

using the euler forward scheme

$$u_E(z_K, t_n) = u_K^n$$

$$u_E(z_{K+1}, t_n) = u_{K+1}^n$$ → $K+1$ = one step deeper from surface

$$u_E(z_{K-1}, t_n) = u_{K-1}^n$$ → one step shallower towards surface

same time level 'n' : the same applies to v

$$\frac{\partial^2 u_E}{\partial z^2} \approx \frac{u_{K+1}^n - 2u_K^n + u_{K-1}^n}{\Delta z^2}$$

$$\frac{\partial^2 v_E}{\partial z^2} \approx \frac{v_{K+1}^n - 2v_K^n + v_{K-1}^n}{\Delta z^2}$$

$$\therefore \frac{u_K^{n+1} - u_K^n}{\Delta t} - f v_K^n = A_v \frac{u_{K+1}^n - 2u_K^n + u_{K-1}^n}{\Delta z^2}$$

$$\therefore \frac{v_K^{n+1} - v_K^n}{\Delta t} - f u_K^n = A_v \frac{v_{K+1}^n - 2v_K^n + v_{K-1}^n}{\Delta z^2}$$

Slide 146 said we solutions $\hat{u_K}$ & $\hat{v_K}$ $\therefore$ we can solve for the next time step by multiplying both sides by Δt

$$u_K^{\hat{n}+1} = u_K^{\hat{n}} + \Delta t \left[f v_K^{\hat{n}} + A_v \frac{u_{K+1}^{\hat{n}} - 2u_K^{\hat{n}} + u_{K-1}^{\hat{n}}}{\Delta z^2} \right]$$

$$v_K^{\hat{n}+1} = v_K^{\hat{n}} + \Delta t \left[-f u_K^{\hat{n}} + A_v \frac{v_{K+1}^{\hat{n}} - 2v_K^{\hat{n}} + v_{K-1}^{\hat{n}}}{\Delta z^2} \right]$$

Question 2

$$\frac{u_K^{\hat{n}+1} - u_K^{\hat{n}}}{\Delta t} = A \frac{u_{K+1}^{\hat{n}} - 2u_K^{\hat{n}} + u_{K-1}^{\hat{n}}}{\Delta z^2} + f v_K^{\hat{n}}$$

$$D = K \frac{\Delta t}{\Delta x^2} \leq \frac{1}{2} \xrightarrow{K \to A_v,\ \Delta x \to \Delta z} D = A_v \frac{\Delta t}{\Delta z^2} \leq \frac{1}{2}$$

From exercise 12, the diffusion stability number was $D = K \frac{\Delta t}{\Delta x^2}$. I replace K with the vertical eddy viscosity A_v & Δx with Δz, since the diffusion is now in the vertical direction

$$\therefore D = A_v \frac{\Delta t}{\Delta z^2} \leq \frac{1}{2}$$

this is a stability condition, if D exceeds 0.5, this will produce unrealistic velocities (oscillating), just like Question 3 of exercise 12 when K was doubled.

Question 3

$\longrightarrow$ vertical (diapycnal) edd viscosity

$A_V = 10^{-2}\ m^2 s^{-1}$ (slide 138 table)

$\left.\begin{array}{l}\Delta z = 2\ m \\ \Delta t = 160\ s\end{array}\right\}$ side 146

$$D = A_v \frac{\Delta t}{\Delta z^2} = 10^{-2}\ \frac{160}{2^2} = 0.4$$

$\hookrightarrow$ below the stability limit of 0.5 $\therefore$ chosen values are stable.

$D = 0.4$

$\therefore \Delta z = 2\ m$ & $\Delta t = 160\ s$ is stable.

$\downarrow$

With $\Delta z = 2m$ over a total depth of $H = 100\ m$, we have 50 vertical grid points

$$\Delta t \leq \frac{0.5 \times \Delta z^2}{A_v} = \frac{0.5 \times 2^2}{10^{-2}} = 200\ s$$

$\therefore$ with $\Delta z = 2\ m$ & $A_v = 10^{-2}\ m^2 s^{-1}$, the time step must be at most $200\ s$ to remain stable.